



Academic year 2022-2023 (ODD Sem)

DEPARTMENT OF INDUSTRIAL ENGINEERING & MANAGEMENT

Date	18 th January 2023	Maximum Marks	10 + 50
Course Code	21IM33	Duration	20 + 90 Min
Sem	III Semester	CIE - I	
WORK SYSTEMS DESIGN			

Note:

1. Answer all the Questions.

PART - A			
Sl. No.	Questions	M	BT CO
1	What is meant by the term Work?	02	2 1
2	Illustrate the Pyramidal Structure of the task.	02	3 1
3	How is Motion Study different from methods engineering?	02	2 2
4	What is meant by productivity?	02	2 1
5	List the three categories of principles of motion economy.	02	1 2

PART - B			
Sl. No.	Questions	M	BT CO
1.	Analyze the problems encountered by organizations in measuring productivity.	10	4 CO1
2.	A work group of 5 workers in a certain month produced 500 units of output working 8 hr/ day for 22 days in the month. What productivity measures could be used for this situation? Suppose that in the next month, the same work group produced 600 units but there were only 20 work days in the month. Using the same productivity measures as before, determine the productivity index using the prior month as a base.	10	3 CO1
3.	Enumerate the principles of motion economy related to the use of Human body.	10	2 CO2
4.	Methods engineering can be categorized into Methods analysis and Methods design. Distinguish between them in terms of their scope, objectives and applications.	10	3 CO2
5.	The normal time to perform a repetitive manual task is 4.25 minutes. In addition, an irregular work element whose normal time is 1.75 minutes must be performed every 8 cycles. Two work units are produced each cycle. The PFD allowance is 16%. Determine the standard time per piece. How many work units are produced in an 8 hour shift at standard performance? Determine the anticipated amount of time worked and the amount of time lost per 8 hour shift that corresponds to the PFD allowance factor of 16%.	10	3 CO3

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution		Particulars	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test		06	04	--	--	--	02	06	02	--	--	--
	Quiz	Max Marks	20	20	10	--	--	--	10	30	10	--	--

Academic year 2022-2023 (Odd Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING AND MANAGEMENT

Date	19 th January 2023	Maximum Marks	10+50
Course Code	21IM34	Duration	20+90 Min
Sem	III Semester	CIE - I	
MANUFACTURING PROCESS			

1. Answer all the Questions.

PART - A				
QNo.1	Questions	M	BT	CO
1.1	In _____ forging operation, repeated hammering and closed die is used.	01	L2	CO1
1.2	_____ Molding machine gives better packing of sand near pattern	01	L1	CO2
1.3	In sand molding, the bottom most part of the flask is called _____	01	L2	CO1
1.4	Over hang core is also called as _____	01	L1	CO2
1.5	_____ Pattern are generally used for circular castings	01	L2	CO1
1.6	Irregular depressions on the surface of the forging is the _____ defect in forging	01	L3	CO2
1.7	Define porosity?	02	L1	CO1
1.8	List any two limitations of Extrusion process	02	L2	CO2

PART - B				
Sl. No.	Questions	M	BT	CO
1.	Define pattern? List the types of patterns and explain any two briefly.	10	L2	CO2
2.	"Sand slinger moulding machine is used for preparing the moulds at faster rate and with high quality". Explain with a neat sketch.	10	L2	CO1
3.	"Molding processes to produce internal cavities and reentrant angles". What is the process called and explain the different types.	10	L3	CO2
4.	Explain the stages in drop forging a connecting rod for an internal combustion engine	10	L3	CO1
5.	Briefly explain the principle of Rolling with a neat sketch	10	L3	CO1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	05	05	--	--	04	05	01	--	--	--
	Test	30	20	--	--	10	20	20	--	--	--



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Department of Industrial Engineering and Management
CONTINUOUS INTERNAL EVALUATION I - September 2019
III SEMESTER

METROLOGY AND MEASUREMENTS - 18IM33

MAX. MARKS: 50

DURATION: 90 min

Course Outcomes:

CO1:	Discuss the principles and practices of metrology in manufacturing environment and analyze uncertainty in an appropriate manner.
CO2:	Describe the operating principles of range of widely used instrumentation techniques and illustrate how to use them in the design of measurement systems.
CO3:	Compare the production process, the product function and the product design, and to select appropriate measurement quantities and tools for these purposes.
CO4:	Evaluate and respond to the need for rigorous and formal metrology concepts in designing and using measurement systems

Q. No	Questions	M	CO	BT
Q1.	With an example, enumerate on the generalized measurement system. Justify the answer by considering bourdon tube as an example. Describe with neat sketches and block diagrams.	(10)	L2	CO1
Q2a	Describe briefly the different static characteristics of Measurement systems.	(06)	L2	CO2
b	Define Error. Explain briefly the different types of Errors and remedies to resolve them.	(04)	L2	CO2
Q3a	Define (i) Maximum Clearance (ii) Basic Shaft (iii) Fundamental Deviation and schematically represent it.	(06)	L1	CO1
b	Enumerate on the concept (i) Interchangeability (ii) Hole basis systems	(04)	L2	CO2
Q4.	Determine the tolerances on the hole and the shaft for a precision running fit designated by 25H7g6, given; 25 mm lies between 18-30 mm - Fundamental deviation for g shaft = $-16D^{0.44}$ - IT7=16i and IT6=10i State the actual maximum and minimum sizes of the hole and shaft and maximum and minimum clearances and represent all the dimensions graphically.	(10)	L4	CO2
Q5a	Define Fit. Explain briefly the various types of fits with examples.	(10)	L2	CO2



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CONTINUOUS INTERNAL EVALUATION - I – Sept. 2019

III SEMESTER

18IM35 – WORK SYSTEMS DESIGN

DURATION: 90 min

MAX. MARKS:50

Instructions to Students:

1. Answer all questions from Part B.

Course Outcomes:

CO1	State the industrial engineering principles that influence the productivity improvement in organizations.
CO2	Apply the methods engineering and operational analysis in re-designing of work systems.
CO3	Apply engineering work measurement principles in analyzing and measurement of work.
CO4	Analyze the work processes using advanced work study tools and techniques.
CO5	Demonstrate an understanding of emerging concepts and applications in designing work systems.

Part – B**Max. Marks 50**

Q.No	Questions	M	BT	CO
Q1 a	Distinguish between Labour productivity ratio and Labour Productivity Index.	(05)	L2	CO1
b	Distinguish between work system as a physical entity and work system as a profession.	(05)	L2	CO1
Q2	A work group of 5 workers in a certain month produced 500 units of output working 8 hr/day for 22 days in the month. (i) What productivity measures could be used for this situation, and what are the values of their respective productivity ratios? (ii) Suppose that in the next month, the same work group produced 600 units but there were only 20 work days in the month. Using the same productivity measures as before, determine the productivity index using the prior month as a base.	(10)	L3	CO1
Q3	The normal time to perform a certain manual work cycle is 3.47 min. In addition, an irregular work element whose normal time is 3.70 min must be performed every 10 cycles. One work unit is produced each cycle. The PFD allowance factor is 14%. Determine (i) the standard time per piece and (b) how many work units are produced during an 8-hour shift at 100% performance, and the worker works exactly 7.018 hr, which corresponds to the 14% allowance factor. (iii) If the worker's pace is 120% and he works 7.2 hours during the regular shift, how many units are produced?	(10)	L4	CO3
Q4	Distinguish between the following along with an example for each. (i) Office automation vs. Office augmentation (ii) Office activities vs. Office applications	(05) (05)	L2	CO1
Q5	Review any five principles related to the design of tooling and equipment.	(10)	L4	CO2

M - Marks, BT - Blooms Taxonomy level, CO - Course outcome



90

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CONTINUOUS INTERNAL EVALUATION- I - September 2019

III SEMESTER

18IM36 – Manufacturing Process

DURATION: 90 min

MAX. MARKS: 50

Instructions to Students:

1. Answer all questions from Part B.

Course outcomes:

CO1	Explain the basic principles and methodology of various manufacturing processes that are used for the production of mechanical parts and products.
CO2	Compare and contrast the advantages and limitations of different manufacturing processes.
CO3	Solve the problems on processing time and economics of processing of material with respect to a manufacturing process.
CO4	Apply the design concept of various manufacturing processes when a specific product has to be manufactured.

Part – B**Max. Marks 50**

Part - B		Max. Marks									
Q.No	Questions	M	BT	CO							
Q1	Discuss orthogonal & oblique cutting with respect to the following: i. Angle of contact ii. Depth of cut iii. Chip form iv. Area of contact	(10)	2	2							
Q2	State the assumptions made by Merchant while constructing a Merchant Circle diagram.	(10)	2	2							
Q3	In an orthogonal turning operations on a lathe the following data were obtained: Cutting force = 120 kg, Feed force = 30 kg, Back rake angle = 15° , Feed rate = 0.2 mm/rev, chip thickness = 0.3 mm, Cutting speed = 100 m/min, Work piece diameter = 120 mm, depth of cut = 0.4 mm. Calculate chip thickness ratio, shear angle, coefficient of friction, friction angle, shear stress, shear strain, strain energy and chip flow velocity.	(10)	1	1							
Q4	In a tool wear test with a high speed steel cutting tool, the following values of tool life were obtained: <table border="1" style="margin-left: auto; margin-right: auto;"> <thead> <tr> <th>Tool Life</th> <th>Cutting Speed</th> </tr> </thead> <tbody> <tr> <td>30 min</td> <td>25.0 m/min</td> </tr> <tr> <td>1.5 min</td> <td>70.0 m/min</td> </tr> </tbody> </table> Calculate the values of 'n' & 'c' using Taylor's Tool Life equation.	Tool Life	Cutting Speed	30 min	25.0 m/min	1.5 min	70.0 m/min	(10)	1	1	
Tool Life	Cutting Speed										
30 min	25.0 m/min										
1.5 min	70.0 m/min										
Q5	Discuss the factors which affect the parameters of Tool Life.	(10)	2	2							

Note: M - Marks, BT - Blooms Taxonomy Level, CO - Course Outcome



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Academic year 2022-2023 (ODD Sem)

**DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT**

Date	22 nd February 2023	Maximum Marks	10 + 50
Course Code	21IM33	Duration	20 + 90 Min
Sem	III Semester	CIE – II	
WORK SYSTEMS DESIGN			

Note:

1. Answer all the Questions.

PART - A				
Sl. No.	Questions	M	BT	CO
1	Distinguish between snap back and cumulative timing methods.	02	2	3
2	According to Taiichi Ohno, the number of waste forms in a manufacturing environment is _____.	01	1	2
3	The meaning of the word Muda in Japanese is _____.	01	1	2
4	Distinguish between accuracy and precision.	02	2	3
5	Distinguish between First generation and Higher generation levels of PMTS.	02	2	3
6	List the five steps in the direct time study procedure.	02	1	3

PART - B																		
Sl. No.	Questions	M	BT	CO														
1.	Classify and explain briefly different methods used to determine the time standards.	10	3	3														
2.	What is a Work Element? Mention at least six guidelines for defining work elements in direct time study. Supplement with appropriate explanation and examples.	10	3	3														
3.	The continuous timing method in direct time study was used to obtain the elemental times for a worker machine task as indicated. Element <i>c</i> is a machine controlled and the time is constant. Elements <i>a, b, d, e</i> and <i>f</i> are operator controlled and external to machine cycle and they are performance rated at 80%. If the machine allowance is 25% and worker PFD allowance is 15%, determine the normal time and standard time for the work cycle. <table><tr><td>Element</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr><tr><td>Observed Time (min)</td><td>0.18</td><td>0.30</td><td>0.88</td><td>1.12</td><td>1.55</td><td>1.80</td></tr></table>	Element	a	b	c	d	e	f	Observed Time (min)	0.18	0.30	0.88	1.12	1.55	1.80	10	3	3
Element	a	b	c	d	e	f												
Observed Time (min)	0.18	0.30	0.88	1.12	1.55	1.80												
4.	Enumerate the three activity sequence models in Basic MOST with suitable examples.	10	4	3														
5.	Illustrate the structure of Lean Production System. Explain the features of different elements of the structure.	10	3	2														

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
	Test	Max Marks	--	10	40	--	--	--	--	40	10	--	--
	Quiz		--	06	04	--	--	04	06	--	--	--	--



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CONTINUOUS INTERNAL EVALUATION - II – October 2019

III SEMESTER

18IM35 – WORK SYSTEMS DESIGN

DURATION: 90 min

MAX. MARKS:60

Instructions to Students:

1. Answer all questions from Part A and B.
2. Use of t distribution tables permitted

Course Outcomes:

CO1	State the industrial engineering principles that influence the productivity improvement in organizations.
CO2	Apply the methods engineering and operational analysis in re-designing of work systems.
CO3	Apply engineering work measurement principles in analyzing and measurement of work.
CO4	Analyze the work processes using advanced work study tools and techniques.
CO5	Demonstrate an understanding of emerging concepts and applications in designing work systems.

Part -A

Max Marks: 10

Q. No	Questions	M	BT	CO
Q1	1.1 Distinguish between the terms "Accuracy" and "Precision".	2	2	1
	1.2 List the four prerequisites for valid time standards.	2	1	3
	1.3 The standard time is referred to as allowed time. Justify.	2	3	3
	1.4 The evaluation of the workers performance is _____.	1	1	3
	1.5 The time units in MTM are called as _____.	1	1	4
	1.6 The motion aggregates in Basic MOST are called as _____.	1	1	4
	1.7 What is the meaning of 'Coverage' in the application of Standard Data System?	1	2	4

Part – B

Max. Marks 50

Q.No	Questions	M	BT	CO															
Q1	The observed element times and performance ratings collected in a direct time study are indicated in the table below. The snapback timing method was used. The PFD allowance in the plant is 14%. All elements are regular elements in the work cycle. Determine (a) the normal time and (b) the standard time for the cycle. <table border="1"> <thead> <tr> <th>Work element</th><th>a</th><th>b</th><th>c</th><th>d</th></tr> </thead> <tbody> <tr> <td>Observed time (min)</td><td>0.22</td><td>0.41</td><td>0.30</td><td>0.37</td></tr> <tr> <td>Performance ratings (%)</td><td>90</td><td>120</td><td>100</td><td>90</td></tr> </tbody> </table>	Work element	a	b	c	d	Observed time (min)	0.22	0.41	0.30	0.37	Performance ratings (%)	90	120	100	90	10	3	3
Work element	a	b	c	d															
Observed time (min)	0.22	0.41	0.30	0.37															
Performance ratings (%)	90	120	100	90															
Q2	The snapback timing method was used to obtain average times for work elements in one work cycle. The times are given in the table below. Element <i>d</i> is a machine-controlled element and the time is constant. Element <i>a, b, c, e</i> and <i>f</i> are operator-controlled and were performance rated at 80%; however, elements <i>e</i> and <i>f</i> are performed during the machine controlled element. The machine allowance is zero (no extra time is added to the machine cycle), and the PFD allowance is 14%. Determine (a) the normal time for the cycle and (b) the standard time for the cycle. <table border="1"> <thead> <tr> <th>Element</th><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr> </thead> <tbody> <tr> <td>Observed time (min)</td><td>0.24</td><td>0.30</td><td>0.17</td><td>0.76</td><td>0.26</td><td>0.14</td></tr> </tbody> </table>	Element	a	b	c	d	e	f	Observed time (min)	0.24	0.30	0.17	0.76	0.26	0.14	10	4	3	
Element	a	b	c	d	e	f													
Observed time (min)	0.24	0.30	0.17	0.76	0.26	0.14													

Name: _____

USN _____

Q3 a.	Seven cycles have been observed during a direct time study. The mean for the largest element time is 0.85 min, and the corresponding sample standard deviation s is 0.15 min, which was also the largest. If the analyst wants to be 95% confident that the mean of the sample was within $\pm 0.10\%$ of the true mean, how many more observations should be taken?	05	3	1
b.	Review the key features of MOST.			
Q4	Illustrate the classification of methods to determine time standards. Explain any two of them in detail.	05	2	4
Q5	Elaborate the procedure in determining time standard using the technique of Predetermined Motion Time Systems(PMTS)	10	3	1
	*****	10	2	3

M - Marks, BT - Blooms Taxonomy level, CO - Course outcome

Q.N	
1.	
2.	
3.	
4. a	
b	



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Department of Industrial Engineering and Management
CONTINUOUS INTERNAL EVALUATION II - October 2019
III SEMESTER
METROLOGY AND MEASUREMENTS - 18IM34

DURATION: 120 min

MAX. MARKS: 60

Instructions to Students:

1. Answer all questions from Part A and Part B.
2. Part A should be written on starting two pages of the answer booklet only.

Course Outcomes:

CO1:	Discuss the principles and practices of metrology in manufacturing environment and analyze uncertainty in an appropriate manner.
CO2:	Describe the operating principles of range of widely used instrumentation techniques and illustrate how to use them in the design of measurement systems.
CO3:	Compare the production process, the product function and the product design, and to select appropriate measurement quantities and tools for these purposes.
CO4:	Evaluate and respond to the need for rigorous and formal metrology concepts in designing and using measurement systems

PART - A

Q. No	Questions	M	BT	CO
1.1	_____ is a new standard in communication used in CAM	(01)	L2	CO3
1.2	_____ is the sensory part of a CMM responsible for sensing different parameters required for measurement.	(01)	L1	CO3
1.3	The data-collecting unit in a CMM is the _____	(01)	L1	CO3
1.4	A pointed instrument used as an input device in the probe of a CMM is _____	(01)	L1	CO3
1.5	List the different types of gauges according to the form of testing surface.	(01)	L1	CO2
1.6	List any two considerations in the selection of gauge material	(01)	L1	CO2
1.7	Wear allowance is provided on the _____ side of the gauge.	(01)	L1	CO2
1.8	'GO' Limit refers to _____	(01)	L1	CO2
1.9	The Optical flat is made up of _____ material	(01)	L2	CO2
1.10	Type B Optical Flat is used for _____	(01)	L2	CO3

PART - B

Q. No	Questions	M
Q1.	Design the 'General' type 'GO' and 'NOGO' gauges for the components having $50H_{8}c_{8}$ fit. Given the following with usual notations. (i) Upper deviation for Shaft = $(95+0.8D)$ microns (ii) 50mm falls in the diameter step of 50-80mm (iii) IT8 = 25i Take wear allowance as 10% of the gauge Tolerance. Also mention the type of fit. Represent the dimensions graphically.	(10)
Q2.	Enumerate on the different types of Main structure of a Coordinate Measuring Machine with neat sketches, and list out the industrial Applications of a CMM	(10)
Q3a	Describe the role of CMM in Computer Aided Manufacturing.	(06)
b.	Explain briefly the advantages of CMM	(04)
Q4.	Describe briefly the Inductive and Optical Transmission Probes with neat sketches.	(10)
Q5a	Explain briefly the Taylor's principle of gauge design with neat sketches.	(06)
b	Enumerate on the Allocation of Tolerance on Gauge Design	(04)

**Instructions to Students:**

1. Answer all questions from Part A in first two pages.
2. Answer all questions from Part B.

Course outcomes:

CO1	Explain the basic principles and methodology of various manufacturing processes that are used for the production of mechanical parts and products.
CO2	Compare and contrast the advantages and limitations of different manufacturing processes.
CO3	Solve the problems on processing time and economics of processing of material with respect to a manufacturing process.
CO4	Apply the design concept of various manufacturing processes when a specific product has to be manufactured.

Part – A**Max. Marks 10**

Q.No	Questions	M	BT	CO
1.	During milling operation cutter moves _____ to its axis.	(01)	1	1
2.	An arbor is used hold the milling _____.	(01)	1	1
3.	In a milling operation, length of the cut is 360 mm and feed rate is 180 mm/min, then milling time in minutes is _____.	(01)	1	1
4.	The use of coatings on milling cutters will increase the hardness of the tool.	(01)	1	1
5.	Gang milling refers to the use of two or more milling cutters mounted on the arbor in a _____ milling set up.	(01)	1	1
6.	Cutting T-slots can be performed more effectively by _____ milling machine.	(01)	1	1
7.	The dividing heads are of _____ types.	(01)	1	1
8.	By linking the worm and worm gear driving mechanism of the attachment with the leadscrew of the table, continuous _____ motion can be imparted to the workpiece.	(01)	1	1
9.	If drill is 5.5 mm, metal removal rate is 65 mm ³ /minute and rpm is 130. Then _____ is value of feed in mm/revolution.	(01)	2	2
10.	If $t=2$ minute, $n=2$ rpm and $s=5$ mm. t =machining time in minute, l =length of travel of the drill in mm, n = rpm of the drill and s =feed per revolution of the drill in mm. Then _____ is the value of l .	(01)	2	2

Part – B**Max. Marks 50**

Q.No	Questions	M	BT	CO
Q1	Explain the nomenclature of a milling cutter with a neat sketch.	(10)	1	1
Q2	Differentiate between up milling & down milling with respect to the following: i. Sketch ii. Alternate name iii. Chip thickness iv. Job lifting tendency v. Practicability	(10)	1	1
Q3	Discuss with a neat sketch the geometry of twist drill.	(10)	1	1
Q4	State clearly, the Plate No. and the indexing movements to be made	(10)	2	2

Name: _____

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using a Universal dividing head, to divide the periphery of a work into 69 equal divisions. Following indexing plates are available:

Plate No. 1:	15	16	17	18	19	20
Plate No. 2:	21	23	27	29	31	33
Plate No. 3:	37	38	41	43	47	49

Q5

A hole with 40 mm diameter and 50 mm depth is to be drilled in mild steel component. The cutting speed can be taken as 65 m/min and the feed rate as 0.25 mm/rev. Calculate the machining time and the metal removal rate.

(10)

Note: M - Marks, BT - Blooms Taxonomy Level, CO - Course Outcome

Name:

USN

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Department of Industrial Engineering and Management
CONTINUOUS INTERNAL EVALUATION - III – Nov. 2019

III SEMESTER

18IM35 – WORK SYSTEMS DESIGN

DURATION: 90 min

MAX. MARKS:50

Instructions to Students:

1. Answer all questions from Part B.

Course Outcomes:

CO1	State the industrial engineering principles that influence the productivity improvement in organizations.
CO2	Apply the methods engineering and operational analysis in re-designing of work systems.
CO3	Apply engineering work measurement principles in analyzing and measurement of work.
CO4	Analyze the work processes using advanced work study tools and techniques.
CO5	Demonstrate an understanding of emerging concepts and applications in designing work systems.

Part – B

Max. Marks 50

Q.No	Questions	M	BT	CO
Q1	Illustrate the role of personal computers and personal digital assistants in computerizing the steps for direct time study.	10	3	4
Q2	Evaluate the role of expert systems in computerized work measurement systems.	10	4	4
Q3	Elaborate the possible contributions to Learning, by the organizations as a part of their continuous improvement strategy.	10	2	1
Q4	Illustrate the structure of a Lean Production System. Elaborate on the features of two pillars in the structure.	10	3	4
Q5	In Toyota Production System a standardized work procedure for a given task has three components. What are they? Explain any two in detail.	10	2	4



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Department of Industrial Engineering and Management
CONTINUOUS INTERNAL EVALUATION III – November 2019
III SEMESTER

METROLOGY AND MEASUREMENTS - 18IM34

MAX. MARKS: 50

DURATION: 90 min

Course Outcomes:

CO1:	Discuss the principles and practices of metrology in manufacturing environment and analyze uncertainty in an appropriate manner.
CO2:	Describe the operating principles of range of widely used instrumentation techniques and illustrate how to use them in the design of measurement systems.
CO3:	Compare the production process, the product function and the product design, and to select appropriate measurement quantities and tools for these purposes.
CO4:	Evaluate and respond to the need for rigorous and formal metrology concepts in designing and using measurement systems

Questions		M	CO	BT
Q. No				
Q1.	Explain briefly the Operating principle of a Autocollimator and describe how it can be used for the measurement of straightness.	(10)	L3	CO4
Q2.	Explain briefly the working principle of a sine bar and with neat sketches, describe the uses of it.	(10)	L3	CO3
Q3a	With neat sketches explain briefly any three systems of displacement amplification used in Mechanical Comparators.	(06)	L2	CO3
b.	Explain the basic principle of a Pneumatic comparator.	(04)	L2	CO4
Q4.	“LVDT works on the principle of Mutual Induction” - Explain briefly its working with neat sketches and list any four applications of it.	(10)	L3	CO3
Q5.	Explain the basic principle of LASER. List the various uses of LASER in metrology and explain any two in detail.	10	L2	CO4

Name: _____

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 Department of Industrial Engineering and Management
 CONTINUOUS INTERNAL EVALUATION- III - Nov 2019
 III SEMESTER

18IM36 - Manufacturing Process
 DURATION: 120 min

MAX. MARKS: 60

Instructions to Students:

1. Answer all questions from Part A in first two pages.
2. Answer all questions from Part B.

Course outcomes:

CO1	Explain the basic principles and methodology of various manufacturing processes that are used for the production of mechanical parts and products.
CO2	Compare and contrast the advantages and limitations of different manufacturing processes.
CO3	Solve the problems on processing time and economics of processing of material with respect to a manufacturing process.
CO4	Apply the design concept of various manufacturing processes when a specific product has to be manufactured.

Part - B**Max. Marks 50**

Q.No	Questions	M	BT	CO
Q1	How is skeleton of the bus being manufactured? Explain briefly with the help of the sketch.	(10)	4	2
Q2	What properties determine the quality of a sand mold for sand casting?	(10)	2	2
Q3	Draw the neat sketch of tool layout for producing a bushes in a capstan/turret lathe.	(10)	4	2
Q4	"Moulding processes to produce internal cavities and reentrant angles". What is the process called and explain the different types.	(10)	2	1
Q5	What are general defects encountered in casting process?	(10)	1	1

Note: M - Marks, BT - Blooms Taxonomy Level, CO - Course Outcome

Academic year 2022-2023 (Odd Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING AND MANAGEMENT

	21 st March 2023	Maximum Marks	10+50
se Code	21IM34	Duration	20+90 Min
	III Semester	Improvement Test	

MANUFACTURING PROCESS

1. Answer all the Questions.

PART A

No.1	Questions	M	BT	CO
1.1	In chemical machining is material removal happens with the help of _____	01	L2	CO3
1.2	In the Electro Chemical Machining process, there is the tool and work piece. Which is the Cathode in the configuration?	01	L1	CO4
1.3	List any two features of advanced machining process	02	L2	CO3
1.4	In AJM, which of the materials are used as abrasive grains?	02	L1	CO2
1.5	Define Hybrid Machining processes	02	L2	CO3
1.6	Summarize difference between EBM and LBM	02	L3	CO4

PART B

Sl. No.	Questions	M	BT	CO
1.	Classify the different types of advanced Machining Processes used in the Industries	10	L2	CO3
2.	Draw the schematic diagram and explain the principle of operation of electrochemical grinding process.	10	L2	CO2
3.	With the aid of sketch, explain the working principle of Water-jet Machining.	10	L3	CO4
4.	Describe the working principle and construction of laser jet machining.	10	L2	CO3
5.	State the differences between additive and subtractive manufacturing	10	L3	CO4

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Quiz	Max	-	02	05	03	02	06	02	-	-	-
	Test	Marks	-	10	20	20	-	30	20	-	-	-

Academic year 2022-2023 (ODD Sem)

DEPARTMENT OF
INDUSTRIAL ENGINEERING & MANAGEMENT

Date	21 st March 2023	Maximum Marks	10 + 50
Course Code	21IM33	Duration	20 + 90 Min
Sem	III Semester	CIE - Improvement	

WORK SYSTEMS DESIGN

Note:

1. Answer all the Questions.

PART - A				
Sl. No.	Questions	M	BT	CO
1	Standard Time is also known as Allowed Time. Justify.	02	2	3
2	Distinguish between First generation and Higher generation levels of PMTS.	02	2	3
3	List the prerequisites for valid time standards.	02	1	3
4	What are the broad two components of Environment in the context and scope of Ergonomics analysis?	02	1	4
5	Classify the three basic categories of Human-Machine systems.	02	2	4

PART - B																		
Sl. No.	Questions	M	BT	CO														
1.	Review different types of Allowances used in Time Standards. Justify the need to consider the same in determining standard times.	10	4	3														
2.	The observed average time in a direct time study was 2.40 minutes for a repetitive work cycle. The worker's performance was rated at 110% on all cycles. The personal time, fatigue and delay allowance for this work is 12%. Determine the normal time and the standard time for the cycle.	10	3	3														
3.	The snap back timing method in direct time study was used to obtain the elemental times for a worker machine task as indicated. Element <i>c</i> is a machine controlled and the time is constant. Elements <i>a, b, d, e</i> and <i>f</i> are operator controlled and external to machine cycle and they are performance rated at 80%. If the machine allowance is 25% and worker PFD allowance is 15%, determine the normal time and standard time for the work cycle. <table border="1"><tr><td>Element</td><td>a</td><td>b</td><td>c</td><td>d</td><td>e</td><td>f</td></tr><tr><td>Observed Time (min)</td><td>0.18</td><td>0.30</td><td>0.88</td><td>1.12</td><td>1.55</td><td>1.80</td></tr></table>	Element	a	b	c	d	e	f	Observed Time (min)	0.18	0.30	0.88	1.12	1.55	1.80	10	4	3
Element	a	b	c	d	e	f												
Observed Time (min)	0.18	0.30	0.88	1.12	1.55	1.80												
4.	Illustrate and Explain through a block diagram model, the typical interactions in a human-machine system.	10	3	4														
5.	Distinguish between the terms, Hazard and Accident. Also illustrate a model depicting the possible factors that contribute to industrial accidents.	10	3	4														

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution		Particulars	CO1	CO2	CO3	CO4	CO5	L1	L2	L3	L4	L5	L6
		Test	--	--	30	20	--	--	--	30	20	--	--
		Quiz	--	--	06	04	--	04	06	--	--	--	--